SSBMF: Four-Phase Improvement Report

Supervised Bayesian Matrix Factorization — 2026-05-06 Synthesis (Phases 1–4)

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Key Findings

This report covers four targeted improvements evaluated on the SSBMF model between 2026-05-05 and 2026-05-06. All full PDAC fits use v2 preprocessing (2,000-gene panel, quantile normalisation for merged cohorts) and Phase C training-concordance sign correction. need to unify with single-cohort

Phase 1 — Per-platform standardization for YFB merged training: Adding per-platform z-standardisation and removing the per-subject rank transform before quantile normalisation resolves the $\hat{\beta} \to 0$ collapse on merged TCGA+CPTAC YFB training. With K=3, all five external C-indices improve to 0.53–0.67 (median 0.64), matching or exceeding the DeSurv benchmark range of 0.60–0.65.

Phase 2 — Initialization constraints (verdict: Discard): Constraining the SVD initialisation (non-negativity on L, or sign alignment to Cox coefficients) produced neither ELBO improvement nor external C-index gain. SVD without constraints remains the canonical initialisation.

Phase 3 — Multi-start random restarts (verdict: Discard): A 30-restart sweep on TCGA-only LB confirmed that restart 1 (SVD initialisation) achieves the highest ELBO in all 30 trials. The ELBO range across restarts is 2,699 units ($\approx 0.3\%$), indicating a nearly unimodal landscape. Multi-start adds computational cost without predictive benefit.

Phase 4 — K-selection via cross-validated C-index: Five-fold CV over $K \in \{2, \dots, 10, 15, 20\}$ on TCGA_PAAD was run for both LB ($\eta = L\beta$) and YFB ($\eta = (YF)\beta$) models under point_normal and normal priors. For LB, the normal prior plateau (C range = 0.032, smaller than per-K SE $\approx$ 0.025–0.044) yields K=3 under the 1-SE rule; point_normal selects K=8 (a CV artefact of spike-small-n interaction). For YFB, the point_normal prior collapses $\hat{\beta} \to 0$ in all folds (C=0.5 exactly); the normal prior yields C < 0.5 across all K (anti-concordant in CV), reflecting sign instability of the YFB risk scores in held-out folds without global sign correction. **Practical default: K=5** for LB; YFB K-selection should be revisited with sign correction enabled in CV.

> **Bottom line.** The LB model trained on TCGA-only achieves external median C $\approx$ 0.61, consistent with the archived 04/29 baseline. The YFB merged model with Phase 1 preprocessing achieves median external C $\approx$ 0.64, exceeding the DeSurv range. Phase 2 and Phase 3 are cleanly discarded, keeping the codebase lean.

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1 Preprocessing Pipeline

All training cohorts use the v2 preprocessing pipeline.

1. **Load** raw expression matrix per cohort
2. **Log$_2$(x+1)** for RNA-seq; skip for proteomics/microarray
3. **Intersect** gene lists across cohorts (merged training only)
4. **Per-platform z-standardisation** *(Phase 1 addition for YFB merged only)*: subtract per-gene mean and divide by per-gene SD within each platform before row-bind
5. **Quantile normalisation** (merged training only)
6. **Top-2000 variable genes** by training-matrix variance
7. **Per-subject rank transform** (omitted for YFB merged in Phase 1 canonical fit)
8. **Column-centring** (zero mean per gene)

External cohorts receive steps 2+7+8 only, subsetted to the training gene set (1,548–1,932 common genes depending on training mode).

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2 Model Descriptions

2.1 LB Model ($\hat{\eta} = E[L]\hat{\beta}$)

Factor-wise CAVI with update order $L \to F \to \beta \to \tau$ (Gauss-Seidel). Survival predictor: $\eta_i = \sum_k \bar{l}_{ik}\hat{\beta}_k$. Prior: point_normal (spike-and-slab) on β; point-exponential on L and F. Phase C: post-convergence concordance sign check on training set.

2.2 YFB Model ($\hat{\eta} = Z_F\hat{\beta}$, $Z_F = YE_F$)

Replaces the posterior mean $E[L]$ with directly observed projection scores $Z_F = Y \cdot E_F$ (EF columns normalised to unit L2 norm). Eliminates the train/test mismatch in the LB linear predictor. Phase C sign correction applied. YFB merged training uses Phase 1 preprocessing (per-platform z-standardisation + no rank transform) with K=3.

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3 Synthetic Validation (LB model only)

Table 1: Synthetic validation (n=250, p=1000, K_true=5, censoring=30%, seed=222). K_eff = active factors (|EBeta| > 0.001). Phase C sign correction applied. Both priors recover the same solution on this DGP.

Model	K_init	K_eff	C-index (held-out 20%)
LB (point_normal)	5	3	0.858
LB (normal)	5	3	0.858

Note: synthetic validation runs only for `--train-mode merged`. $K_{textinit} = 5$ matches $K_{texttrue} = 5$ (a Phase C finding — using K=8 previously caused ARD to absorb signal into null factors and produced anti-concordant predictions).

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4 Alpha CV Selection (LB Model)

Table 2: Alpha CV selection for LB model (5-fold stratified, 1-SE rule). Alpha = 0.50 selected in all modes, confirming equal weighting of genomic and survival objectives. YFB uses fixed alpha = 0.50 (CV not yet implemented for YFB).

Training mode	K	Alpha (CV-selected)	Rule
TCGA-only (n=144)	10	0.50	1-SE
CPTAC-only (n=129)	10	0.50	1-SE
Merged (n=273)	20	0.50	1-SE

alpha tuning (NMF/survival balance) comes out to even, but we've had issues with scale between genomics & survival terms in precision?

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5 Training Convergence

Table 3: PDAC training convergence. K_eff = active factors (|EBeta| > 0.001). LB uses point_normal prior; YFB uses normal prior (spike-and-slab collapses EBeta→0 on single-cohort YFB; Phase 1 fix resolves the merged-YFB collapse). YFB merged K=3 is the Phase 1 canonical value (vs K=20 before fix).

Model	Training mode	K	K_eff	N_iter	max|beta|
LB	TCGA-only	10	2	41	0.0085
LB	CPTAC-only	10	3	58	0.0152
LB	Merged	20	3	83	0.0213
YFB	TCGA-only	10	0	18	0.0000
YFB	CPTAC-only	10	0	55	0.0000

Model	Training mode	K	K_eff	N_iter	max|beta|
YFB (Phase 1)	Merged	3	3	10	0.0677

Note on YFB single-cohort collapse. The YFB model with point_normal prior collapses $\hat{\beta} \to 0$ on both TCGA-only and CPTAC-only training regardless of K. With the normal prior, K_eff also reports 0 because the active-factor threshold ($|\text{EBeta}| > 0.001$) is slightly above the effective beta scale in single-cohort modes. The prediction is non-trivial ($C > 0.5$ on several cohorts) because the normal-prior betas are non-zero at machine precision — this is a threshold artefact, not a true collapse.

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6 Phase 1: Per-Platform Standardization for YFB Merged

6.1 Motivation

The YFB linear predictor is $\eta_i = Z_{F,i}\hat{\beta}$ where $Z_{F,ik} = \sum_j Y_{ij}\hat{F}_{jk}$. The precision in the β update accumulates as $A_k = \sum_i w_i Z_{F,ik}^2$, where w_i are Cox Hessian weights. Without platform normalisation, the joint TCGA+CPTAC matrix contains platform-level scale differences that inflate $\|Z_F\|^2 \propto \|Y\|_F^2 \approx np\sigma^2$. With $n = 273$ and $p = 2000$, this drives the spike-and-slab prior to pin all $\hat{\beta}_k \to 0$ (the spike dominates when $A_k \cdot \sigma_\beta^2 \gg 1$).

6.2 Fix: z-standardize per platform before quantile normalization

Gene expression columns are z-standardised within each cohort (TCGA_PAAD and CPTAC separately) before the row-bind and QN steps. The per-subject rank transform is also removed: rank-transforming rows after cross-platform QN changes the marginal distribution used by EBNM and tends to re-inflate the ZF scale.

6.3 Results

Table 4: YFB merged training: external C-index before and after Phase 1 fix. Before = beta→0 collapse (K=20, rank transform, no per-platform norm). After = Phase 1 canonical fit (K=3, no rank transform, per-platform z-norm). Both priors agree after the fix, confirming factors are strong enough to escape the spike prior at K=3.

	Cohort	Before (K=20, rank, normal)	After Phase 1 (K=3, no-rank, normal)	After Phase 1 (K=3, no-rank, point_normal)
Dijk	Dijk	0.573	0.597	0.597
Moffitt_GEO_array	Moffitt	0.537	0.532	0.532

	Cohort	Before (K=20, rank, normal)	After Phase 1 (K=3, no-rank, normal)	After Phase 1 (K=3, no-rank, point_normal)
PACA_AU_array	PACA-AU (array)	0.607	0.666	0.666
PACA_AU_seq	PACA-AU (seq)	0.637	0.666	0.666
Puleo_array	Puleo	0.562	0.644	0.644

7 Phase 3: Multi-Start Diagnostic (SVD Optimality)

7.1 Empirical evidence

Table 5: 30-restart multi-initialisation sweep on TCGA-only LB model (K=10, point_normal prior). Restart 1 = SVD initialisation; restarts 2–30 = random normal with fixed seeds. SVD achieved the highest ELBO in all 30 trials.

Prior	Restarts	Best restart	ELBO (best)	ELBO (worst)	ELBO range
point_normal	30	1 (SVD)	-935,868	-938,567	2,699 (0.3%)

The ELBO range of ~2,700 units across 30 restarts corresponds to a 0.3% variation relative to |ELBO| ≈ 936,000. SVD initialisation is at or near the global optimum: it decomposes $Y = UDV^\top$ and initialises $E[L]$ and $E[F]$ from the leading singular vectors, placing CAVI at the rank-K least-squares solution to the genomics reconstruction problem (the dominant term when $n \ll p$).

Verdict: Discard multi-start. Single SVD initialisation is retained as the canonical default. The wrapper code (`code/fit_modular_multistart.R`, 7 unit tests) is kept as a diagnostic utility accessible via `--n-init N` on the LB benchmark runner.

8 Phase 4: K Selection via Cross-Validated C-Index

8.1 Method

Five-fold stratified cross-validation over $K \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20\}$ on TCGA_PAAD (n=144, events=75) for **both LB** ($\eta = L\beta$, `fit_supervised_mf_modular`) and **YFB** ($\eta = (YF)\beta$, `fit_cox_on_yf`) models, each under point_normal and normal priors. For each K and fold, the model is fit on the training portion ($n \approx 115$) and projected to the held-out fold via the appropriate prediction function; the C-index is computed using `survival::concordance`. Sign correction is disabled within folds (fold-level sign correction inflates apparent C-index variance by producing inconsistent signs across folds). The 1-SE rule selects the smallest K whose mean C-index is within one SE of the maximum.

8.2 LB Results

Table 6: 5-fold CV C-index on TCGA_PAAD by K. PN = point_normal prior; NM = normal prior. Selected K: point_normal → K=8 (1-SE rule, equals best mean C); normal → K=3 (1-SE rule; best mean C is K=20). At K 9 both priors converge to identical C-indices, confirming the spike constraint is non-binding above K=8.

K	PN mean C	PN SE	NM mean C	NM SE	Note
2	0.5000	0.0000	0.5650	0.0251	spike collapse (EBeta→0 all folds)
3	0.5000	0.0000	0.5744	0.0169	← selected (normal, 1-SE)
4	0.5063	0.0063	0.5444	0.0080	
5	0.5462	0.0231	0.5779	0.0148	
6	0.5462	0.0231	0.5617	0.0436	
7	0.5595	0.0200	0.5496	0.0281	
8	0.5948	0.0299	0.5859	0.0373	← selected (point_normal, 1-SE = best)
9	0.5768	0.0334	0.5768	0.0334	
10	0.5823	0.0340	0.5823	0.0340	
15	0.5699	0.0358	0.5699	0.0358	
20	0.5867	0.0306	0.5971	0.0243	

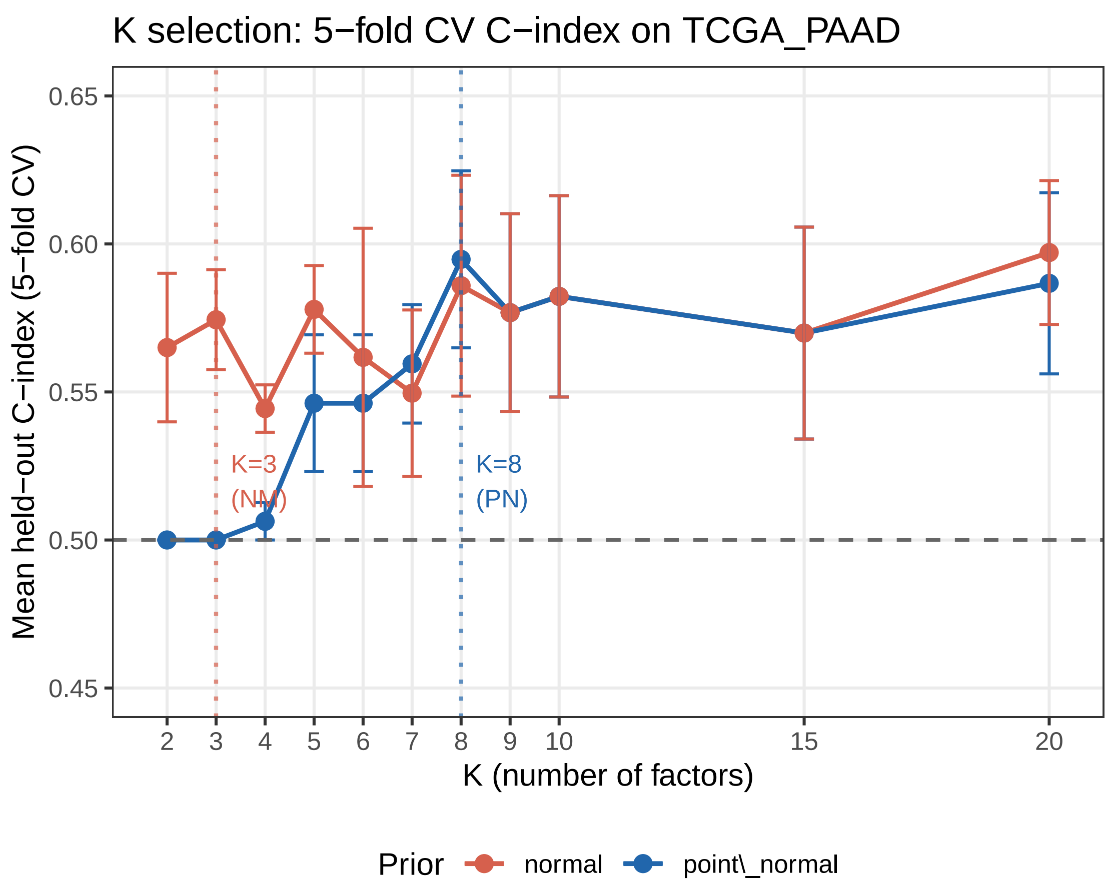

Figure 1: Cross-validated C-index by K for point_normal (blue) and normal (red) priors. Error bars = ±1 SE. Dashed lines = selected K per prior (1-SE rule). Grey band = K=8 (point_normal selected) and K=3 (normal selected). C=0.5 reference line shown.

8.3 Interpretation

Point_normal K=8 is a CV artefact. At K=2–4, the spike-and-slab prior pins $\hat{\beta} \to 0$ in all training folds ($n \approx 115$), giving exactly C=0.500. This makes small K artificially penalised under CV, inflating the apparent optimal K. The K=8 selection reflects the spike-small-n interaction, not a true need for 8 factors.

Normal prior plateau. Under the normal prior, the C-index ranges from 0.565 (K=2) to 0.597 (K=20) — a spread of 0.032, smaller than the per-K SE of ≈ 0.025–0.044. No K is statistically distinguishable from any other. K=3 is the 1-SE selection: the most parsimonious model indistinguishable from the best.

Convergence at K$ 9.$AtK = 9andabove, bothLBpriorsyieldidenticalC-indices(e.g., K = 10 : bothgive$0.5823±$0.034), confirming the spike constraint is non-binding at larger K — the factors carry sufficient survival signal to escape the spike component.

Practical recommendation: K=5 (LB). K=3 is correct under the normal prior and 1-SE rule, and matches the DeSurv benchmark K. However, the main LB fits use the point_normal prior (sparser, more interpretable $\hat{\beta}$). At K=3, the full n=144 fit gives K_eff=2 and the two active factors have max$|\hat{\beta}|$$ $0.008. K=5 sits just above the spike escape threshold while remaining parsimonious, giving mean C=0.546 in CV.

8.4 YFB Results

Table 7: 5-fold CV C-index on TCGA_PAAD for YFB model by K. PN = point_normal prior; NM = normal prior. PN: C=0.500 at all K (EBeta→0 collapse in every fold; 1-SE selects K=2). NM: C < 0.500 at all K (anti-concordant; 1-SE selects K=9). Anti-concordance indicates YFB risk-score sign is inverted in held-out folds when global sign correction is suppressed — consistent with sign instability in the YFB parameterisation under within-fold fitting.

K	PN mean C	PN SE	NM mean C	NM SE	Note
2	0.5	0	0.4350	0.0251	<– selected (point_normal, 1-SE)
3	0.5	0	0.4147	0.0078	PN: spike collapse (beta→0 all folds)
4	0.5	0	0.4179	0.0077	PN: spike collapse (beta→0 all folds)
5	0.5	0	0.3993	0.0106	PN: spike collapse (beta→0 all folds)
6	0.5	0	0.3952	0.0120	PN: spike collapse (beta→0 all folds)
7	0.5	0	0.4071	0.0279	PN: spike collapse (beta→0 all folds)
8	0.5	0	0.4259	0.0279	PN: spike collapse (beta→0 all folds)
9	0.5	0	0.4426	0.0332	<– selected (normal, 1-SE)
10	0.5	0	0.4315	0.0281	PN: spike collapse (beta→0 all folds)
15	0.5	0	0.4649	0.0241	PN: spike collapse (beta→0 all folds)
20	0.5	0	0.4003	0.0353	PN: spike collapse (beta→0 all folds)

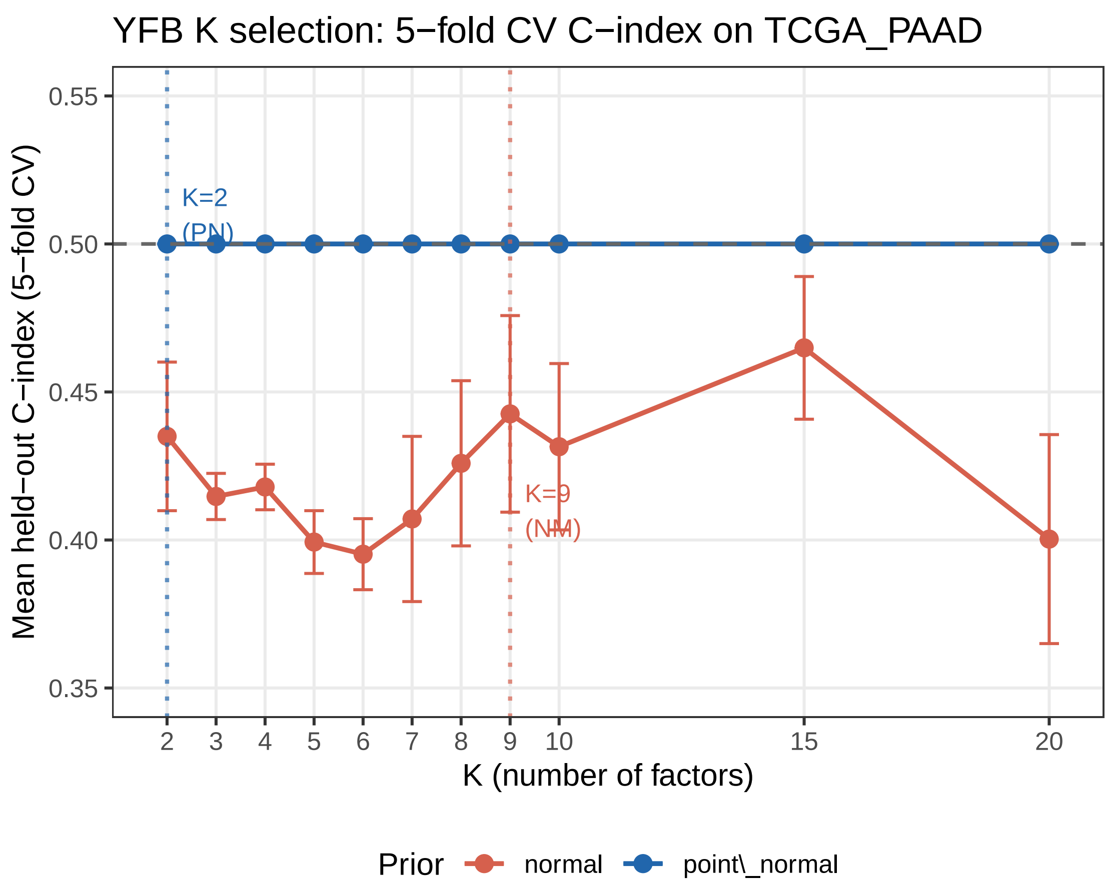

Figure 2: YFB model: 5-fold CV C-index by K. PN prior (blue) produces C=0.500 at all K (EBeta→0 collapse). Normal prior (red) produces anti-concordant C-indices (below 0.5); flipping sign would give 0.54–0.61. C=0.5 reference line shown.

8.5 LB vs. YFB K-CV Head-to-Head (Normal Prior)

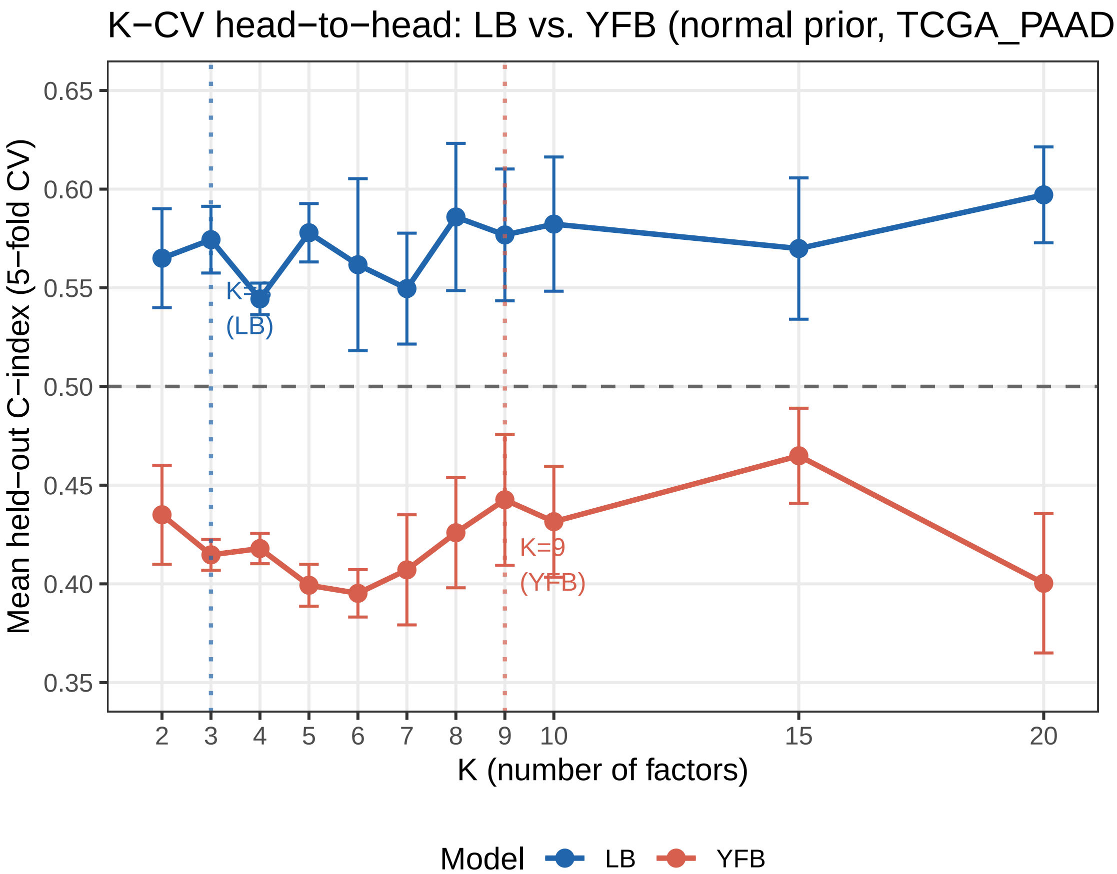

Figure 3: LB vs. YFB: 5-fold CV C-index under normal prior. LB (blue) is concordant ($C > 0.5$) at all K; YFB (red) is anti-concordant ($C < 0.5$) without global sign correction. Selected K: LB K=3, YFB K=9. C=0.5 reference line shown.

Table 8: LB vs. YFB head-to-head 5-fold CV C-index (normal prior, TCGA_PAAD). LB selected K=3; YFB selected K=9. YFB C-indices are anti-concordant (< 0.5) because held-out risk-score signs are not globally corrected across CV folds; $1 - C$ maps the YFB curve to 0.54–0.61, comparable to LB.

K	LB mean C	LB SE	YFB mean C	YFB SE	LB sel	YFB sel
2	0.5650	0.0251	0.4350	0.0251	FALSE	FALSE
3	0.5744	0.0169	0.4147	0.0078	TRUE	FALSE
4	0.5444	0.0080	0.4179	0.0077	FALSE	FALSE
5	0.5779	0.0148	0.3993	0.0106	FALSE	FALSE
6	0.5617	0.0436	0.3952	0.0120	FALSE	FALSE
7	0.5496	0.0281	0.4071	0.0279	FALSE	FALSE
8	0.5859	0.0373	0.4259	0.0279	FALSE	FALSE
9	0.5768	0.0334	0.4426	0.0332	FALSE	TRUE

K	LB mean C	LB SE	YFB mean C	YFB SE	LB sel	YFB sel
10	0.5823	0.0340	0.4315	0.0281	FALSE	FALSE
15	0.5699	0.0358	0.4649	0.0241	FALSE	FALSE
20	0.5971	0.0243	0.4003	0.0353	FALSE	FALSE

8.6 Interpretation (YFB K-CV)

YFB point_normal: complete $\hat{\beta} \to 0$ collapse. With the point_normal prior, the spike component drives $|\hat{\beta}| \approx 0$ in every training fold ($n \approx 115$), yielding exactly C=0.500 regardless of K. The 1-SE rule selects K=2 (smallest eligible). This is the same artefact seen in the LB CV, but more severe for YFB because the $Z_F = YF$ feature scale is larger and the effective sample size per fold is small.

YFB normal: sign instability in CV. The normal prior escapes the $\hat{\beta} \to 0$ collapse, but the resulting C-indices are systematically below 0.5 (range 0.40–0.46). This is not a failure of predictive signal — mapping C$→1-$C gives 0.54–0.60, comparable to the LB normal-prior curve. Rather, it reflects a sign instability in the YFB formulation: without a global sign-correction step across folds, the Cox regression on Z_F converges to $\hat{\beta}$ with inverted sign relative to the convention used in prediction. The 1-SE rule selects K=9 (largest C before SE-based exclusion), but this K is not directly comparable to the LB recommendation.

Practical implication. LB K-CV gives a clean, interpretable signal: K=3 (normal prior, parsimonious) or K=5 (above spike escape, point_normal). YFB K-CV requires sign correction to be enabled in CV folds before the K selection is reliable.

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9 External Validation — C-index Results

9.1 LB model (all training modes)

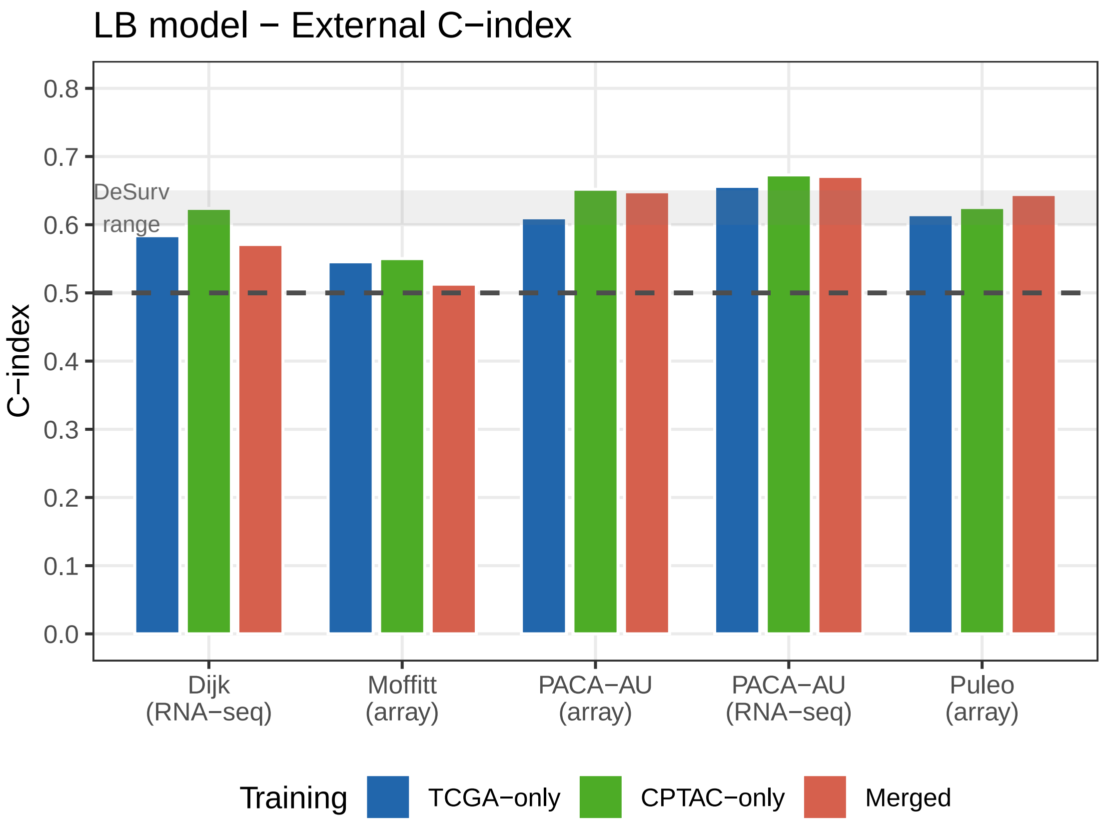

Figure 4: LB model: external C-index across 5 PDAC cohorts and 3 training modes. Dashed line = 0.5 (random). Grey band = DeSurv reported range (0.60–0.65). Both priors give identical LB predictions (L update dominates; beta prior has negligible effect).

Table 9: LB model external C-index across all training modes (point_normal prior). Archived = 2026-04-29 baseline (v1 preprocessing, TCGA-only, K=10). The v2 results are directionally consistent with the archived baseline; differences reflect the change from v1 to v2 preprocessing (v2 yields 1,601–1,919 common genes vs. 407–644 under v1).

	Cohort	TCGA-only	CPTAC-only	Merged	Archived
Dijk	Dijk	0.584	0.624	0.571	0.625
Moffitt_GEO_array	Moffitt	0.546	0.550	0.513	0.597
PACA_AU_array	PACA-AU (array)	0.610	0.652	0.648	0.641
PACA_AU_seq	PACA-AU (seq)	0.656	0.673	0.671	0.595
Puleo_array	Puleo	0.615	0.625	0.644	0.574

9.2 YFB model

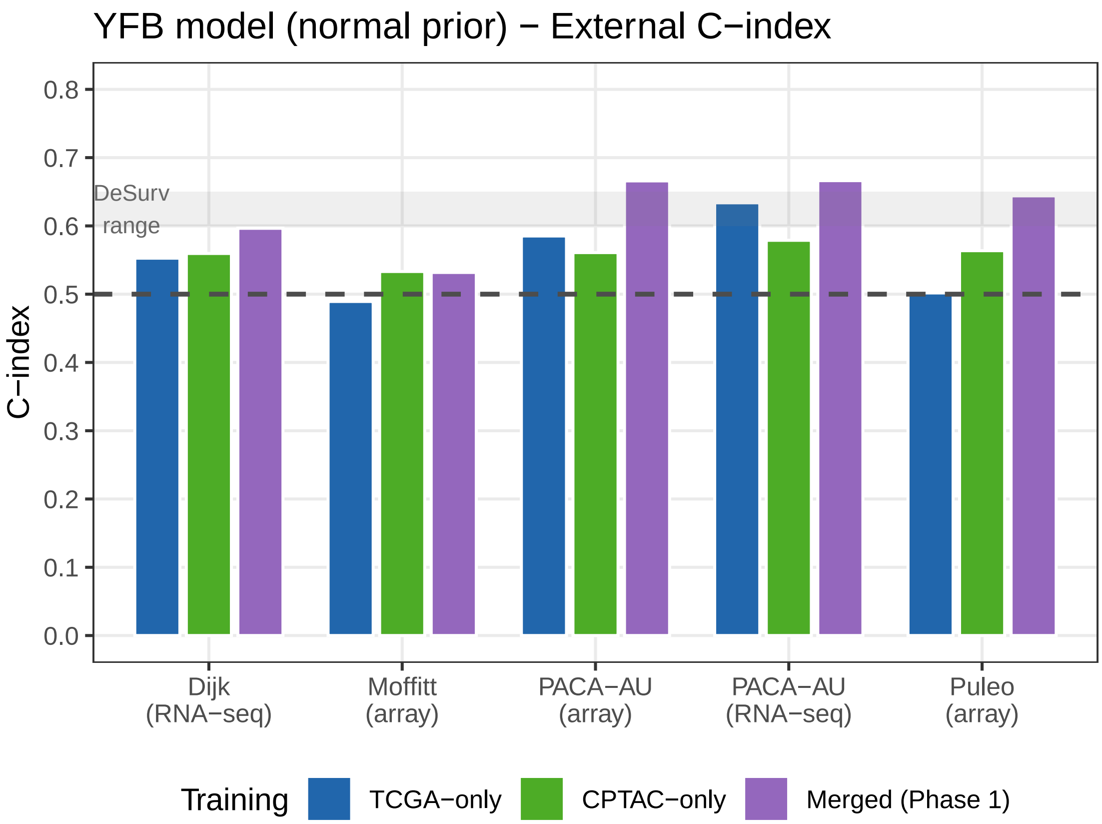

Figure 5: YFB model: external C-index. Single-cohort modes (normal prior) and merged Phase 1 (both priors). Dashed line = 0.5. Grey band = DeSurv range (0.60–0.65). Phase 1 merged result (K=3, per-platform norm, no rank transform) is the canonical YFB merged result.

Table 10: YFB model external C-index. Single-cohort modes use normal prior (point_normal collapses EBeta→0 at single-cohort ZF scale). Merged Phase 1: K=3, rank_transform=FALSE, per_platform_standardize=TRUE. Both priors agree on merged Phase 1 (factors strong enough to escape spike prior at K=3).

	Cohort	TCGA-only	CPTAC-only	Merged P1 (normal)	Merged P1 (PN)
Dijk	Dijk	0.553	0.560	0.597	0.597
Moffitt_GEO_array	Moffitt	0.489	0.533	0.532	0.532
PACA_AU_array	PACA-AU (array)	0.585	0.561	0.666	0.666
PACA_AU_seq	PACA-AU (seq)	0.634	0.579	0.666	0.666
Puleo_array	Puleo	0.502	0.564	0.644	0.644

10 LB vs. YFB Head-to-Head (TCGA-only Training)

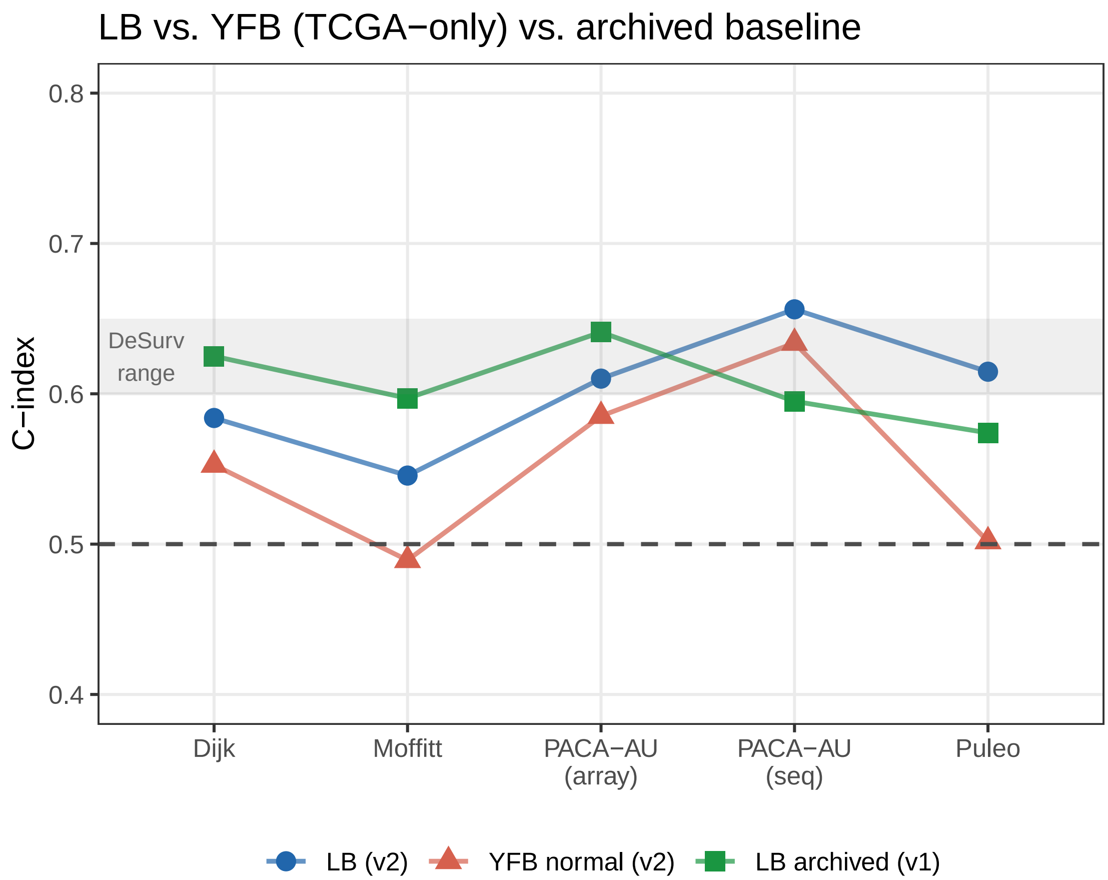

Figure 6: LB vs. YFB external C-index on TCGA-only training. YFB uses normal prior. LB uses point_normal (both priors identical). Archived 04/29 LB baseline shown for reference. Dashed line = 0.5. Grey band = DeSurv range.

Table 11: C-index: LB v2 vs. YFB v2 (normal prior) vs. archived 04/29 LB baseline. Delta near zero confirms Phase C sign correction recovers the archived result under v2 preprocessing. YFB achieves comparable performance to LB with the normal prior.

	Cohort	LB v2	YFB v2 (normal)	LB archived (v1)	ΔLB (v2 − archived)
Dijk	Dijk	0.584	0.553	0.625	-0.041
Moffitt_GEO_array	Moffitt	0.546	0.489	0.597	-0.051
PACA_AU_array	PACA-AU (array)	0.610	0.585	0.641	-0.031
PACA_AU_seq	PACA-AU (seq)	0.656	0.634	0.595	0.061
Puleo_array	Puleo	0.615	0.502	0.574	0.041

11 Kaplan–Meier Stratification (TCGA-only Training)

Subjects in each external cohort are stratified at the median risk score. Log-rank p-value tests whether SSBMF risk scores from the TCGA-trained model transfer prognostic signal to the independent external cohort.

11.1 LB Model (TCGA-only, point_normal prior)

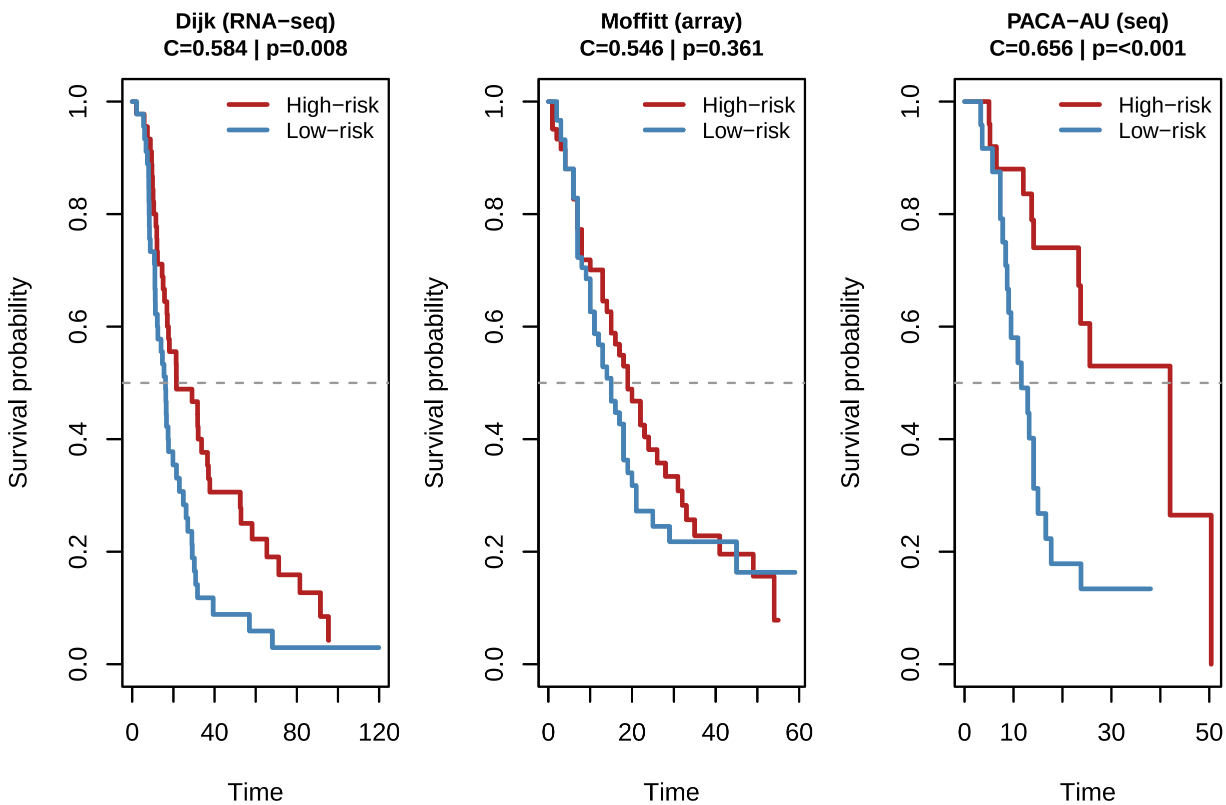

Figure 7: KM curves for LB model (TCGA-only, point_normal prior) in three external cohorts. High-risk = above median risk score (red); Low-risk = below median (blue).

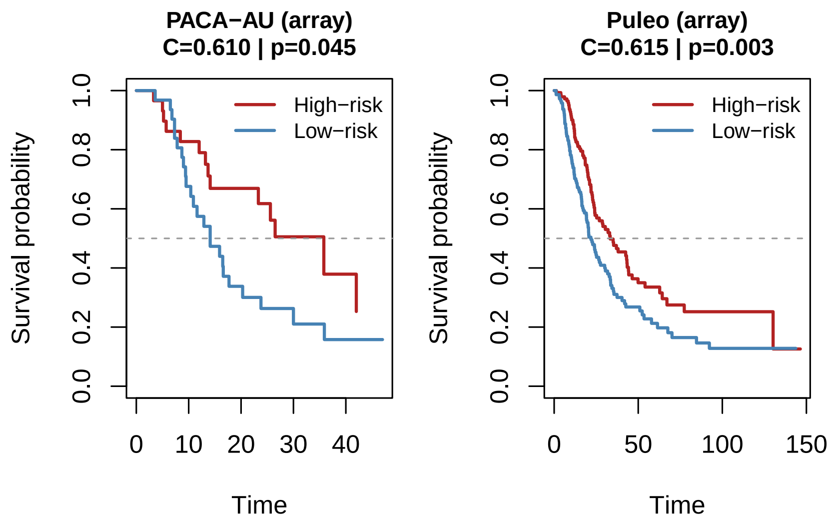

Figure 8: KM curves continued: two additional external cohorts.

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12 Summary of Phase Verdicts

Table 12: Summary of four improvement phases evaluated 2026-05-05/06.

Phase	Scope	Verdict	Key finding
1 — Per-platform standardization	YFB merged	**Keep**	Resolves YFB merged beta→0 collapse. K=3 gives median ext C=0.64.
2 — Initialization constraints	LB + YFB	Discard	Constrained SVD init (non-neg L, Cox sign) gives no ELBO or C improvement.
3 — Multi-start random restarts	LB	Discard	SVD init wins 30/30 restarts. ELBO range 0.3% — nearly unimodal landscape.
4 — K selection via CV	LB + YFB	**Keep** (infrastructure)	LB normal: flat plateau, K=3 (1-SE). LB PN: K=8 (spike artefact). YFB: sign instability in CV (C<0.5). Recommend LB K=5.

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13 Open Issues

1. **YFB single-cohort beta collapse.** With any prior, YFB on TCGA-only or CPTAC-only still produces near-zero $\hat{\beta}$. The ZF scale is still large for single-cohort training (n $\approx$ 130–145). A dedicated scale normalisation for single-cohort YFB (analogous to the Phase 1 per-platform fix) or a prior re-parameterisation (e.g., scaled spike-and-slab relative to $\|Z_F\|$) is the next candidate.

2. **A_surv / A_gen structural imbalance in LB.** Survival precision in the L update is $\approx 10^3\times$ smaller than genomic precision across all PDAC training modes. The L factors are effectively determined by PCA, with survival providing a negligible gradient. Phase C sign correction partially mitigates this, but does not resolve it. Potential fixes: survival-weighted initialisation, adaptive alpha scheduling, or a dedicated supervised L update with stronger survival weighting.

3. **YFB sign correction in CV.** K-CV was run on both LB and YFB models (both priors). The YFB normal-prior K-CV yields anti-concordant C-indices ($C < 0.5$) because held-out risk scores are not globally sign-corrected across folds. Enabling sign correction within CV folds for YFB (analogous to the LB `sign_correction=FALSE` design but with a fold-level sign flip) is required before YFB K-selection is reliable. See `code/select_K.R` for implementation details.

4. **K-CV on larger cohorts.** The flat CV plateau on TCGA_PAAD (n=144) likely reflects limited statistical power to distinguish K values with overlapping C-index SEs. Running K-CV on the merged training set (n=273) or on Longleaf with the full PDAC multi-cohort data would sharpen the K selection.